

5.5 Equilibria (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.5 Equilibria (A Level Only)
Difficulty	Medium

Time allowed: 80

Score: /63

Percentage: /100

Question 1a

i)

State what is meant by partition coefficient.

[2]

Ammonia is soluble in both water and organic solvents.

An aqueous solution of ammonia is shaken with the immiscible organic solvent trichloromethane. The mixture is left to reach equilibrium.

Samples are taken from each layer and titrated with dilute hydrochloric acid.

- A 25.0 cm^3 sample from the trichloromethane layer requires 13.0 cm^3 of $0.100\text{ mol dm}^{-3}\text{ HCl}$ to reach the end-point.
- A 10.0 cm^3 sample from the aqueous layer requires 12.5 cm^3 of $0.100\text{ mol dm}^{-3}\text{ HCl}$ to reach the end-point.

ii)

Calculate the partition coefficient, K_{pc} , of ammonia between trichloromethane and water.

Show your working.

 $K_{\text{pc}} = \dots\dots\dots$

[2]

iii)

Butylamine, $\text{C}_4\text{H}_9\text{NH}_2$, is also soluble in both water and organic solvents.

Suggest how the numerical value of K_{pc} of butylamine between trichloromethane and water would compare to the value of K_{pc} calculated in **(a)(ii)**. Explain your answer.

[2]

[6 marks]

Question 1b

Butanamide, $\text{C}_3\text{H}_7\text{CONH}_2$, is much less basic than butylamine. Explain why.

[1 mark]**Question 2a**

The feasibility of a chemical reaction depends on the standard Gibbs free energy change, ΔG° . This is dependent on the standard enthalpy and entropy changes, and the temperature.

State and explain whether the following processes will lead to an increase or decrease in entropy.

i)
the reaction of magnesium with hydrochloric acid

entropy change

explanation

[1]

ii)
the dissolving of solid potassium chloride in water

entropy change

explanation

[1]

iii)
the condensing of water from steam

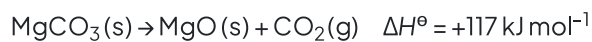
entropy change

explanation

[1]**[3 marks]**

Question 2b

Magnesium carbonate can be decomposed on heating.



Standard entropies are shown in Table 2.1.

Table 2.1

substance	MgCO ₃ (s)	MgO(s)	CO ₂ (g)
S° / J K ⁻¹ mol ⁻¹	+65.7	+26.9	+214

i)

Calculate $\Delta G^\ominus$ for this reaction at 298 K.

Show your working.

$$\Delta G^\ominus = \dots\dots\dots \text{kJ mol}^{-1}$$

[3]

ii)

Explain why this reaction is feasible only at high temperatures.

[1]

[4 marks]

Question 2c

Table 2.2 lists values of solubility products, K_{sp} , of some Group 2 carbonates.

Table 2.2

	solubility product in water at 298 K, $K_{sp} / \text{mol}^2 \text{dm}^{-6}$
MgCO_3	1.0×10^{-5}
CaCO_3	5.0×10^{-9}
SrCO_3	1.1×10^{-10}

Deduce the trend in the solubility of the Group 2 carbonates down the group. Justify your answer using the data given.

[1 mark]

Question 2d

i)
Write an equation to show the equilibrium for the solubility product of MgCO_3 . Include state symbols.

..... =

[1]

ii)
With reference to your equation in **(d)(i)**, suggest what is observed when a few cm^3 of concentrated Na_2CO_3 (aq) are added to a saturated solution of MgCO_3 . Explain your answer.

[2]

[3 marks]

Question 2e

Use the data in Table 2.2 to calculate the solubility of MgCO_3 in water at 298 K, in g dm^{-3} .

Show your working.

solubility of MgCO_3 = g dm^{-3}

[2 marks]

Question 2f

Describe and explain the variation in the thermal stabilities of the carbonates of the Group 2 elements.

[3 marks]

Question 3a

The pH of a $0.150 \text{ mol dm}^{-3}$ solution of piperidine is 11.9.

20.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ HCl was slowly added to a 20.0 cm^3 sample of $0.150 \text{ mol dm}^{-3}$ piperidine. The pH was measured throughout the addition. One mole of hydrochloric acid reacts one mole of piperidine.

i)

Calculate the number of moles of HCl remaining at the end of the addition.

moles of HCl =
[2]

ii)

Hence calculate the $[\text{H}^+]$ and the pH at the end of the addition.

pH =
[1]

[3 marks]

Question 3b

On the following axes shown in Fig. 3.1, sketch how the pH will change during the addition of a total of 20.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ HCl.

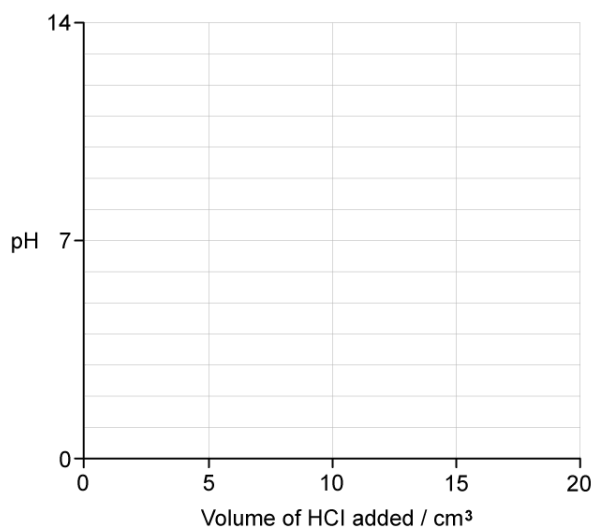


Fig. 3.1

[3 marks]

Question 3c

The student repeated the titration using two different chemicals, 25.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ nitric acid, $\text{HNO}_3 (\text{aq})$, and $0.100 \text{ mol dm}^{-3}$ ammonia, $\text{NH}_3 (\text{aq})$.

i)
State the equation for this reaction.

[1]

ii)
Explain why the salt produced in this reaction is acidic.

[4]

[5 marks]

Question 3d

State the equation for the reaction between nitric acid and water and identify the conjugate acid formed in the reaction.

[2 marks]

Question 4a

This question is about buffers.

Give the meaning of the term buffer solution.

[1 mark]

Question 4b

Buffers can be acidic but they can also be basic. This is determined by the reactants which are combined to create the buffer

i)

Describe how an acidic buffer is made.

ii)

State the key feature which makes a buffer acidic.

[2 marks]

Question 4c

Ethanoic acid is a weak acid. Hydrogen carbonate ions act as a weak acid if they are in an aqueous solution.

i)

Write equations for each of these weak acids at equilibrium.

[2]

ii)

A solution was made up containing sodium hydrogen carbonate and sodium carbonate. Explain how this solution would act as a buffer if a small amount of acid was added to it.

[2]

[4 marks]

Question 4d

Explain how the solution of ethanoic acid works as a buffer when small amounts of alkali are added.

[2 marks]

Question 4e

A buffer solution contains 0.10 mol dm^{-3} concentration of methanoic acid and a 0.20 mol dm^{-3} concentration of sodium methanoate. The values of K_a for methanoic acid is $1.80 \times 10^{-5} \text{ mol dm}^{-3}$.

Calculate the pH of this buffer solution at 298 K. Give your answer to 2 decimal places and show your working.

pH =

[4 marks]

Question 5a

This question is about solubility products

A $1.70 \times 10^{-3} \text{ mol dm}^{-3}$ solution of calcium nitrate is mixed with an equal volume of $1.50 \times 10^{-3} \text{ mol dm}^{-3}$ potassium sulfate.

Predict whether calcium sulfate ($K_{sp} = 2.00 \times 10^{-5}$) will precipitate. Show your working.

[3 marks]

Question 5b

Water with a manganese ion, $\text{Mn}^{2+}(\text{aq})$, concentration above $1.8 \times 10^{-6} \text{ mol dm}^{-3}$ will cause clothes to stain when being washed. The manganese can precipitate as $\text{Mn}(\text{OH})_2(\text{s})$ which has a K_{sp} of 4.5×10^{-14} . This reduces this concentration of $\text{Mn}^{2+}(\text{aq})$ by the addition of OH^{-} ions.

i)

State the expression for K_{sp}

[1]

ii)

Calculate the concentration of OH^{-} ions. Show your working. $[\text{OH}^{-}] = \dots\dots\dots \text{mol dm}^{-3}$
[2]

iii)

Calculate the minimum pH at 298 K required to prevent clothes from being stained.
($K_{\text{w}} = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ (at 298 K (25 °C)))

pH =

[2]

[5 marks]

Question 5c

Consider the following equilibrium containing a saturated solution of AgBr(s)



Explain what will happen to the solubility of AgBr(s) if the following substances are added to the solution.

- AgNO₃

.....

- NaBr

.....

- Pb(NO₃)₂

.....

[3 marks]

Question 5d

Explain why more Mg(OH)₂ dissolves when hydrochloric acid is added to a saturated solution of Mg(OH)₂.

[3 marks]

Model Answers: Medium

1a

a)

i) The partition coefficient is:

- The ratio of the concentrations of a solute in two (immiscible) solvents; [1 mark]
- At equilibrium; [1 mark]

ii) The partition coefficient, K_{pc} , of ammonia between trichloromethane and water is:

- Aqueous layer $[\text{NH}_3] = 0.1 \times \frac{12.5}{10.0} = 0.125 \text{ mol dm}^{-3}$

AND

Trichloromethane layer $[\text{NH}_3] = 0.1 \times \frac{13.0}{25.0} = 0.052 \text{ mol dm}^{-3}$; [1 mark]

- $K_{pc} = \frac{0.052}{0.125} = 0.416$

OR

$$K_{pc} = \frac{0.125}{0.052} = 2.40; [1 \text{ mark}]$$

iii) How the numerical value of K_{pc} of butylamine between trichloromethane and water compares to the value of K_{pc} calculated in **(a)(ii)**:

- If $K_{pc} = 0.416$, the K_{pc} for butylamine will be larger than the K_{pc} for ammonia

OR

If $K_{pc} = 2.40$, the K_{pc} for butylamine will be smaller than the K_{pc} for ammonia; [1 mark]

- Butylamine contains a hydrophobic / non-polar (C_4) chain

AND

Therefore, butylamine is more soluble in trichloromethane; [1 mark]

[Total: 6 marks]

- The definition of partition coefficient is another term that you have to learn
- In part (ii), to calculate the partition coefficient:

1. Write down the equilibrium equation
$$\text{NH}_3(\text{aq}) \rightleftharpoons \text{NH}_3(\text{organic solvent})$$
2. Write down the K_{pc} expression
$$K_{\text{pc}} = \frac{[\text{NH}_3(\text{organic solvent})]}{[\text{NH}_3(\text{aq})]}$$
3. Use the information in the question to determine the concentration of ammonia in the aqueous and organic layer
4. Substitute these values into the expression and evaluate it to get the K_{pc} value of 0.416
 - If you did the calculation with the expression upside down, you will get a final accepted value of 2.40
5. Since the value of K_{pc} is **less than 1**, ammonia is more soluble in water than the organic solvent
 - If you did the calculation with the expression upside down and got a final value of 2.40, this still shows that ammonia is more soluble in water because the value is directly linked to the expression
- In part (iii), you have to consider the differences between ammonia and butylamine
 - The difference is the presence of the butyl chain in butylamine
 - The butyl chain is non-polar / hydrophobic
 - This means that butylamine is more soluble in trichloromethane

1b

b) Butanamide is less basic than butylamine because:

- The carbonyl group (of the amide) means that the lone pair is not available to accept a proton

OR

The electron withdrawing oxygen (of the amide) means that the lone pair is not available to accept a proton; [1 mark]

[Total: 1 mark]

- The basicity of amines and amides depends on the availability of the lone pair of electrons on nitrogen
 - Some functional groups, e.g. alkyl group, donate electron density making the lone pair more available to react
 - Some functional groups, e.g. carbonyl groups, withdraw electron density

making the lone pair less available to react

2a

a) The entropy changes that occur in the following processes are:

i) The reaction of magnesium with hydrochloric acid:

- Entropy increases

AND

(Because,) a gas is formed; [1 mark]

ii) The dissolving of solid potassium chloride in water:

- Entropy increases

AND

(Because,) the ordered solid lattice goes / dissolves to a solution with (free) moving ions; [1 mark]

iii) The condensing of water from steam:

- Entropy decreases

AND

(Because,) the number of moles of gas decreases **OR** (Because,) a more ordered liquid is formed; [1 mark]

[Total: 3 marks]

- **Remember:** Entropy is a measure of disorder

- i) The reaction of magnesium with hydrochloric acid:

- $\text{Mg (s)} + 2\text{HCl (aq)} \rightarrow \text{MgCl}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$
- The reactants and products both have one aqueous chemical
- The reactants include a solid, while the products include a gas
- Gases are more disordered than solids, so there is an increase in entropy

ii) The dissolving of solid potassium chloride in water:

- $\text{KCl (s)} \rightarrow \text{KCl (aq)}$
- Aqueous chemicals are more disordered than solids
 - The ions are unable to move in a solid, but they can flow when they are dissolved in solution

- Therefore, there is an increase in entropy
- iii) The condensing of water from steam:
 - $\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$
 - Gases are more disordered than liquids
 - Gases have lots of energy resulting in the particles moving rapidly and randomly, while liquids are more ordered with particles flowing close together
 - Therefore, there is a decrease in entropy

2b

b)

i) $\Delta G^\ominus$ for this reaction at 298 K is:

- $\Delta S^\ominus = 26.9 + 214 - 65.7 = (+) 175.2 \text{ (J K}^{-1} \text{ mol}^{-1}\text{)}; [1 \text{ mark}]$
- $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$

OR

$$\Delta G^\ominus = 117 - (298 \times \frac{175.2}{1000}); [1 \text{ mark}]$$

- $\Delta G^\ominus = + 64.8 \text{ (kJ mol}^{-1}\text{)}$

AND

Answer given to a minimum of 3 significant figures; [1 mark]

ii) This reaction is only feasible at high temperatures because:

- $T\Delta S^\ominus$ is more positive than $\Delta H^\ominus$

AND

So, $\Delta G^\ominus$ is negative; [1 mark]

[Total: 4 marks]

- **Remember:** Entropy change calculations are basically products – reactants
- **Careful:** One of the most common mistakes for these calculations is to forget to convert the entropy into $\text{kJ K}^{-1} \text{ mol}^{-1}$
 - The units for entropy change are $\text{J K}^{-1} \text{ mol}^{-1}$

- Reactions are feasible when $\Delta G^\ominus \leq 0$
- In this question, $\Delta H^\ominus$ and $\Delta S^\ominus$ are both positive
- This means that T needs to be high for $T\Delta S^\ominus > \Delta H^\ominus$ and cause $\Delta G^\ominus$ to become negative

2c

c) The trend in the solubility of the Group 2 carbonates down the group is:

- The solubility decreases (as you move) down the group

AND

K_{sp} decreases; [1 mark]

[Total: 1 mark]

- Careful:** The values given in Table 2.2 are in standard form and raised to negative powers, which means that the numbers are getting smaller
- This answer can be justified by giving information from Table 2.2, as long there is some form of comparison, e.g.
 - MgCO_3 has a solubility of $1.0 \times 10^{-5} \text{ mol}^2 \text{ dm}^{-6}$ compared to SrCO_3 with a solubility of $1.1 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$
 - MgCO_3 is 2000 times more soluble than CaCO_3

2d

d)

i) The equation to show the equilibrium for the solubility product of MgCO_3 is:

- $\text{MgCO}_3(\text{s}) + (\text{aq}) \rightleftharpoons \text{Mg}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq});$ [1 mark]

ii) When a few cm^3 of concentrated $\text{Na}_2\text{CO}_3(\text{aq})$ are added to a saturated solution of MgCO_3 , you would see:

- A (white) solid / precipitate (of MgCO_3) forming; [1 mark]
- (Because,) an increase in the concentration of carbonate ions, shifts the equilibrium to the left / reactants; [1 mark]

[Total: 3 marks]

- For part (i)

- The + (aq) is used to represent the addition of an excess of water which dissolves the ions into an aqueous solution
- For part (ii)
 - The concentrated sodium carbonate contains sodium ions and carbonate ions
 - The magnesium carbonate is in equilibrium in the magnesium carbonate solution
 - Adding sodium carbonate increases the number of carbonate ions in the solution
 - The equilibrium will shift to the left to reduce the concentration of carbonate ions
 - This results in the formation of solid magnesium carbonate, which will be seen as a solid / precipitate in the solution

2e

e) The solubility of MgCO_3 in water at 298 K, in g dm^{-3} is:

- Solubility in $\text{mol dm}^{-3} = \sqrt{1.0 \times 10^{-5}} = 3.2 \times 10^{-3} \text{ mol dm}^{-3}$; [1 mark]
- Solubility in $\text{g dm}^{-3} = 3.2 \times 10^{-3} \times 84.3 = 0.27 \text{ g dm}^{-3}$; [1 mark]

[Total: 2 marks]

- To complete this calculation:
 1. Write the equilibrium equation
 - $\text{MgCO}_3(\text{s}) \rightleftharpoons \text{Mg}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$
 2. Write the equilibrium expression
 - $K_{\text{sp}} = [\text{Mg}^{2+}(\text{aq})][\text{CO}_3^{2-}(\text{aq})]$
 3. Simplify the equilibrium expression
 - The ratio of $\text{Mg}^{2+}(\text{aq})$ ions : $\text{CO}_3^{2-}(\text{aq})$ ions is 1 : 1
 - $K_{\text{sp}} = [\text{Mg}^{2+}(\text{aq})]^2$
 4. Substitute the values into the expression
 - $1.0 \times 10^{-5} = [\text{Mg}^{2+}(\text{aq})]^2$
 5. This rearranges, simplifies and evaluates to:
 - $[\text{Mg}^{2+}(\text{aq})]^2 = 1.0 \times 10^{-5}$
 - $[\text{Mg}^{2+}(\text{aq})] = \sqrt{1.0 \times 10^{-5}}$
 - $[\text{Mg}^{2+}(\text{aq})] = 3.2 \times 10^{-3} \text{ mol dm}^{-3}$
 - Since the ratio of $\text{MgCO}_3(\text{s})$: $\text{Mg}^{2+}(\text{aq})$ ions is 1 : 1, then the solubility of magnesium carbonate is also $3.2 \times 10^{-3} \text{ mol dm}^{-3}$

- To convert from concentrations in mol dm^{-3} to g dm^{-3} , you multiply by the M_r

2f

f) The variation in the thermal stabilities of the carbonates of the Group 2 elements is:

Chemical ideas expressed:

- The thermal stability increases down the group
- OR**
- More (heat / thermal) energy is required as you move down the group
- As the ionic radius of the cation increases
 - Leading to less distortion / polarisation
- OR**
- Less weakening of (C–O / C=O) bonds
- Of the anion
- OR**
- Of the carbonate / CO_3^{2-} ion

All **four** chemical ideas; [3 marks]

Any **three** chemical ideas; [2 marks]

Any **two** chemical ideas; [1 mark]

[Total: 3 marks]

- The general trend is that the thermal stability increases as you move down the group
 - As you move down Group 2, the ionic radius of the metal cation increases
 - This means that there is more shielding between the positive nucleus and the negative carbonate ion
 - Therefore, the metal cation causes less distortion / polarisation of the negative carbonate ion
 - The amount of energy required to decompose the carbonate ion is linked to the amount of distortion / polarisation
 - More distortion / polarisation = less energy

[Total: 2 marks]

- To complete this calculation:
 1. Write the equilibrium equation
 - $\text{MgCO}_3(\text{s}) \rightleftharpoons \text{Mg}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$
 2. Write the equilibrium expression
 - $K_{\text{sp}} = [\text{Mg}^{2+}(\text{aq})][\text{CO}_3^{2-}(\text{aq})]$
 3. Simplify the equilibrium expression
 - The ratio of $\text{Mg}^{2+}(\text{aq})$ ions : $\text{CO}_3^{2-}(\text{aq})$ ions is 1 : 1
 - $K_{\text{sp}} = [\text{Mg}^{2+}(\text{aq})]^2$
 4. Substitute the values into the expression
 - $1.0 \times 10^{-5} = [\text{Mg}^{2+}(\text{aq})]^2$
 5. This rearranges, simplifies and evaluates to:
 - $[\text{Mg}^{2+}(\text{aq})]^2 = 1.0 \times 10^{-5}$
 - $[\text{Mg}^{2+}(\text{aq})] = \sqrt{1.0 \times 10^{-5}}$
 - $[\text{Mg}^{2+}(\text{aq})] = 3.2 \times 10^{-3} \text{ mol dm}^{-3}$
 - Since the ratio of $\text{MgCO}_3(\text{s})$: $\text{Mg}^{2+}(\text{aq})$ ions is 1 : 1, then the solubility of magnesium carbonate is also $3.2 \times 10^{-3} \text{ mol dm}^{-3}$
- To convert from concentrations in mol dm^{-3} to g dm^{-3} , you multiply by the M_r

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More (heat / thermal) energy is required as you move down the group
- As the ionic radius of the cation increases
- Leading to less distortion / polarisation
OR
Less weakening of (C–O / C=O) bonds
- Of the anion
OR
Of the carbonate / CO_3^{2-} ion

All **four** chemical ideas; [3 marks]

Any **three** chemical ideas; [2 marks]

Any **two** chemical ideas; [1 mark]

[Total: 3 marks]

- The general trend is that the thermal stability increases as you move down the group
 - As you move down Group 2, the ionic radius of the metal cation increases
 - This means that there is more shielding between the positive nucleus and the negative carbonate ion
 - Therefore, the metal cation causes less distortion / polarisation of the negative carbonate ion
 - The amount of energy required to decompose the carbonate ion is linked to the amount of distortion / polarisation
 - More distortion / polarisation = less energy
 - Less distortion / polarisation = more energy

3a

a)

i) The number of moles of HCl remaining at the end of the addition is:

- $n(\text{HCl}) \text{ at start} = \frac{20}{1000} \times 0.200 = 4 \times 10^{-3} \text{ mol}; [1 \text{ mark}]$
- $n(\text{piperidine}) = \frac{20}{1000} \times 0.150 = 3 \times 10^{-3} \text{ mol}; [1 \text{ mark}]$

AND

$$n(\text{HCl}) \text{ at end} = 4 \times 10^{-3} - 3 \times 10^{-3} = 1 \times 10^{-3} \text{ mol}$$

ii) The $[\text{H}^+]$ and the pH at the end of the addition is:

- $\text{conc}(\text{H}^+) = \frac{1 \times 10^{-3}}{40} \times 1000 = 0.025$
 $\text{pH} = -\log 0.025 = 1.60; [1 \text{ mark}]$

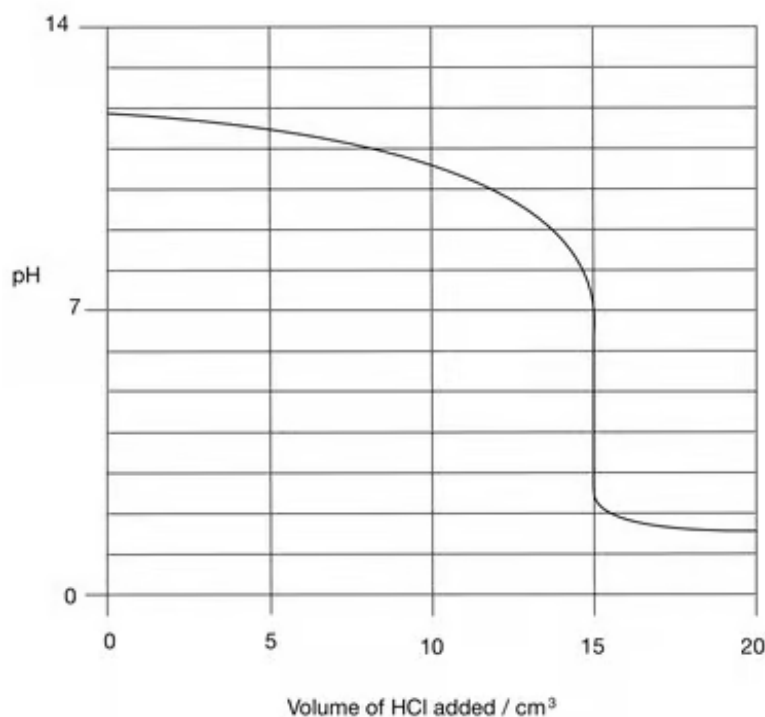
[Total: 3 marks]

- Remember you must work out the number of moles for both HCl and piperidine to find out the moles of HCl remaining
- The HCl is in excess so the solution will have a low pH

- In order to calculate the pH you must have the concentration of H^+ , not the number of moles
 - pH is defined by $-\log[H^+]$

3b

The pH curve should look like:



- pH/vol curve: start at pH 11.9; [1 mark]
- Vertical portion at $V = 15.0 \text{ cm}^3$; [1 mark]
- Levels off at pH 1.6; [1 mark]

[Total: 3 marks]

- You know the starting and ending pH so make sure you mark these on the graph first so you have a reference to work towards as it can be tricky to draw these curves
- You must make sure that the vertical section is in the correct place
- You know that acid is in excess so you can work out the volume of acid required by:
 - $\text{Conc}_{\text{acid}} \times \text{Volume}_{\text{acid}} = \text{Conc}_{\text{base}} \times \text{Volume}_{\text{base}}$
 - Which gives $0.200 \text{ mol dm}^{-3} \times \text{Volume}_{\text{acid}} = 0.150 \text{ mol dm}^{-3} \times 20.0 \text{ cm}^3$

$$\circ \text{ Volume}_{\text{acid}} = \frac{0.150 \text{ mol dm}^{-3} \times 20.0 \text{ cm}^3}{0.200} = 15.0 \text{ cm}^3$$

- You could also be asked to identify the equivalence point of this titration
 - As it is a weak base, strong acid the equivalence point will be below pH 7

3c

c)

i) The equation for the reaction between nitric acid, $\text{HNO}_3(\text{aq})$, and ammonia, $\text{NH}_3(\text{aq})$ is:

- $\text{HNO}_3(\text{aq}) + \text{NH}_3(\text{aq}) \rightarrow \text{NH}_4\text{NO}_3(\text{aq})$; [1 mark]

ii) The salt produced in this reaction is acidic because:

- Ammonium nitrate / $\text{NH}_4\text{NO}_3(\text{aq})$ / dissociates into $\text{NH}_4^+(\text{aq})$ and $\text{NO}_3^-(\text{aq})$; [1 mark]
- Ammonium ion / $\text{NH}_4^+(\text{aq})$ is the conjugate acid of ammonia / $\text{NH}_3(\text{aq})$; [1 mark]
- Ammonium ions / $\text{NH}_4^+(\text{aq})$ react with hydroxide ions / $\text{OH}^-(\text{aq})$

OR



- Acidity of solution increases / pH of solution decreases; [1 mark]

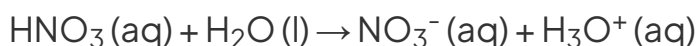
[Total: 5 marks]

- The salt ammonium nitrate / $\text{NH}_4\text{NO}_3(\text{aq})$ will dissociate and form two ions, $\text{NH}_4^+(\text{aq})$ and $\text{NO}_3^-(\text{aq})$
- The positive ion is the ammonium ion, $\text{NH}_4^+(\text{aq})$
- When the positive ion is from a **weak base** then the salt is **acidic** by hydrolysis
- When the negative ion is from a weak acid then the salt is basic by hydrolysis
- Examiners commented that salt hydrolysis was not known well, so by learning the basic rules you will have a good place to start when answering these questions
 - Strong acid - strong base = neutral solution
 - Strong acid - weak base = acidic solution
 - Weak acid - strong base = alkaline solution

- The positive ion is the ammonium ion, NH_4^+ (aq)
- When the positive ion is from a **weak base** then the salt is **acidic** by hydrolysis
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 - Strong acid - strong base = neutral solution
 - Strong acid - weak base = acidic solution
 - Weak acid - strong base = alkaline solution

3d

d) The equation for the reaction between nitric acid and water is:



- Correct species and $\rightarrow$ used; [1 mark]
- The conjugate acid is H_3O^+ ; [1 mark]

[Total: 2 marks]

- Nitric acid, $\text{HNO}_3(\text{aq})$ is a strong acid, therefore the arrow should be $\rightarrow$ and not a reversible sign
- To identify the conjugate acid and base pairings:



Acid	Base	Conj. base	Conjugate acid
------	------	------------	----------------

H^+ donor	H^+ acceptor	H^+ acceptor	H^+ donor
--------------------	-----------------------	-----------------------	--------------------

4a

a) The term buffer solution is:

- (A buffer solution) resists / minimises changes in pH when small amounts of acid or alkali are added; [1 mark]

[Total: 1 mark]

- This is an important definition to know and also to be able to apply having

completed a buffer calculation

4b

b)

i) Describe how an acidic buffer is made:

- By mixing a weak acid with a salt of the same weak acid; [1 mark]

ii) Different buffers have different pH values

- Acidic buffers have a pH of less than / below 7; [1 mark]

[Total: 2 marks]

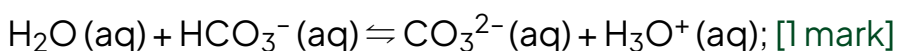
- An example of an acidic buffer is using ethanoic acid with sodium ethanoate:
 - $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$
 - $\text{CH}_3\text{COO}^-\text{Na}^+ \rightleftharpoons \text{CH}_3\text{COO}^- + \text{Na}^+$
 - The salt **must** be one of the **same weak acid**, so that the carboxylate ion (in this case the CH_3COO^-) is the same
 - The salt will fully dissociate, whereas the acid will only partially dissociate
 - For the buffer to work, there must be enough of the carboxylate ion present
- A buffer is a solution that can resist pH change when small amounts of acid or base are added and so maintains the pH
 - This only happens if small amounts of acid or alkali are added

4c

c)

i) The equation for ethanoic acid at equilibrium is:

- $\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$; [1 mark]
- $\text{HCO}_3^-(\text{aq}) \rightleftharpoons \text{CO}_3^{2-}(\text{aq}) + \text{H}^+(\text{aq})$

OR

ii) A solution of sodium hydrogen carbonate and sodium carbonate would act as a buffer if a small amount of acid was added to it by:

- Concentration of H^+ ions would increase / H^+ ions are added

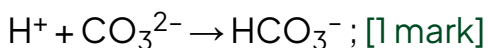
AND

Equilibrium would shift to the left hand side

OR

H^+ ions would react with the carbonate ions in the solution

OR



- The concentration of H^+ ions remains (almost) constant

OR

The additional H^+ ions are removed; [1 mark]

[Total: 4 marks]

- The equation must show that it is an equilibrium reaction not a one way arrow
 - State symbols are not required
- If asked to explain how a buffer solution works, you have the choice between explaining in words or with equations
 - But, make sure you know both how to explain **AND** the equations
 - An exam question could ask you to provide equations to demonstrate what is happening in the buffer

4d

d) The solution of ethanoic acid works as a buffer when small amounts of alkali are added because:

- OH^- / hydroxide ions are added to the buffer solution

AND

The equilibrium moves / shifts to the right / towards the products

OR

OH^- / hydroxide ions react with hydrogen ions / H^+ ; [1 mark]

- An exam question could ask you to provide equations to demonstrate what is happening in the buffer

4d

d) The solution of ethanoic acid works as a buffer when small amounts of alkali are added because:

- OH^- / hydroxide ions are added to the buffer solution

AND

The equilibrium moves / shifts to the right / towards the products

OR

OH^- / hydroxide ions react with hydrogen ions / H^+ ; [1 mark]

- The concentration of hydrogen ions / H^+ remains (almost) constant

OR

The hydrogen ions / H^+ are replaced; [1 mark]

[Total: 2 marks]

- The buffer reacts to changes according to Le Chatelier's Principle.
 - If acid or alkali is added to the buffer solution then the position of equilibrium will move to resist that change
 - You need to be able to explain and provide equations to show what happens in both situations

4e

e) The pH of the buffer is calculated:

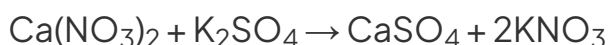
- $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$; [1 mark]
- $[\text{H}^+] = \frac{K_a \times [\text{HA}]}{[\text{A}^-]}$; [1 mark]
- $[\text{H}^+] = \frac{(1.8 \times 10^{-5} \times 0.1)}{0.2} = 9.0 \times 10^{-6}$; [1 mark]
- $\text{pH} = -\log(9 \times 10^{-6}) = 5.05$; [1 mark]

[Total: 4 marks]

- To calculate the pH the equation must be rearranged to find $[H^+]$, from this the pH can be calculated
- In this reaction we can assume that $[A^-]$ is the same as the concentration of sodium methanoate:
 - $HCOONa \rightarrow HCOO^- + Na^+$
 - Since it is a salt, it will fully dissociate
 - So, whatever the concentration of the sodium methanoate is, that is the concentration of the $HCOO^- (A^-)$

5a

a) To determine whether or not calcium sulfate will precipitate:



$$[Ca^{2+}(aq)] = \frac{1.70 \times 10^{-3}}{2} = 8.50 \times 10^{-4} \text{ mol dm}^{-3}$$

AND

$$[SO_4^{2-}(aq)] = \frac{1.50 \times 10^{-3}}{2} = 7.50 \times 10^{-4} \text{ mol dm}^{-3}; [1 \text{ mark}]$$

- $[Ca^{2+}(aq)] \times [SO_4^{2-}(aq)] = 8.50 \times 10^{-4} \times 7.50 \times 10^{-4} = 6.375 \times 10^{-7} / 6.38 \times 10^{-7} \text{ mol}^2 \text{ dm}^{-6}; [1 \text{ mark}]$
- Therefore, there will be no precipitation as Q is less than K_{sp} (of calcium sulfate); [1 mark]

[Total: 3 marks]

- You must halve the concentrations of the solutions as you are doubling the volume
 - The question says you are adding equal volumes of both solutions
- The calculation $[Ca^{2+}(aq)] \times [SO_4^{2-}(aq)]$ will give Q for the reaction
- If this value is greater than K_{sp} then a precipitate will form
- If this value is less than K_{sp} then a precipitate will not form

5b

b)

i) The expression for K_{sp} is:

- $K_{sp} = [\text{Mn}^{2+}][\text{OH}^-]^2$; [1 mark]

ii) To calculate the concentration of OH^-

- $K_{sp} = 1.8 \times 10^{-6} \times [\text{OH}^-]^2$

OR

$$[\text{OH}^-]^2 = 2.5 \times 10^{-8}; \text{ [1 mark]}$$

- $[\text{OH}^-] = 1.58 \times 10^{-4} \text{ (mol dm}^{-3}\text{)}; \text{ [1 mark]}$

iii) To calculate the minimum pH required:

- $K_w = [\text{H}^+][\text{OH}^-] = 1 \times 10^{-14}$

OR

$$[\text{H}^+] = \frac{1.00 \times 10^{-14}}{1.58 \times 10^{-4}} = 6.32 \times 10^{-11}; \text{ [1 mark]}$$

- $\text{pH} = -\log[\text{H}^+] = 10.2; \text{ [1 mark]}$

[Total: 5 marks]

- Careful: 2 moles of OH^- ions are required for this reaction
 - $\text{Mn(OH)}_2(\text{s}) \rightleftharpoons \text{Mn}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$
 - This has an impact on the K_{sp} expression as well as the calculation for $[\text{OH}^-]$
- You can also work the pH out by:
 - $\text{pH} = 14 - (-\log 1.58 \times 10^{-4})$
- Note that the value for K_w is given at the end of the paper in the data table so it wouldn't be given to you in the question

5c

c) On addition of:

- AgNO_3 : Added Ag^+ ions reduce solubility of AgBr as equilibrium shifts to the left; [1 mark]

- If this value is less than K_{sp} then a precipitate will not form

5b

b)

i) The expression for K_{sp} is:

- $K_{sp} = [\text{Mn}^{2+}][\text{OH}^-]^2$; [1 mark]

ii) To calculate the concentration of OH^-

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- Careful: 2 moles of OH^- ions are required for this reaction
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5c

c) On addition of:

- AgNO_3 : Added Ag^+ ions reduce solubility of AgBr as equilibrium shifts to the left; [1 mark]

- NaBr: Adding Br^- ions reduce solubility of AgBr as equilibrium shifts to the left; [1 mark]
- $\text{Pb}(\text{NO}_3)_2$: Adding Pb^{2+} ions forms a precipitate with Br^- so equilibrium shifts to right increasing solubility of AgBr; [1 mark]

[Total: 3 marks]

- When a solution containing a common ion is added the equilibrium shifts to the left hand side and therefore reduces the solubility of the solid which is AgBr in this case
- This is because the increase in the common ion concentration means that $[\text{Ag}^+(\text{aq})][\text{Br}^-(\text{aq})]$ is **greater** than the K_{sp} for AgBr
- As a result, the AgBr is **precipitated**
- Adding Pb^{2+} ions will cause a precipitate with Br^- ions
 - This will remove Br^- ions so the equilibrium will shift to the right hand side

5d

d) The solubility increases because:

- $\text{Mg}(\text{OH})_2(\text{s}) \rightleftharpoons \text{Mg}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$; [1 mark]
- H^+ ions from HCl react with OH^- ions forming H_2O
OR
 $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{H}_2\text{O}(\text{l})$; [1 mark]
- Pushing equilibrium to the right hand side increasing solubility of $\text{Mg}(\text{OH})_2$; [1 mark]

[Total: 3 marks]

- The addition of H^+ ions from a strong acid (fully dissociated) will react with the OH^- ions present
- This removes the OH^- ions and therefore forces the equilibrium to the right hand side
- Therefore the $\text{Mg}(\text{OH})_2$ will become more soluble or the solubility will increase